WEEK 3 · DAY 1 · OVERVIEW

Visualization



Created by **Jeff Lenamon**

ABOUT 60 MIN WITH EXERCISES / 10 SECTIONS / 7 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The Desk Lead's Note

Week 2 gave me numbers: a mean spread of 185.75 bp, 128 bonds investment grade, the book by sector. Numbers hide the shape. Show me how far the spread tail runs, how spreads step up by rating, where the money sits, and what moves together. Title and label every chart; I will forward them.

Your Route Through Day 1

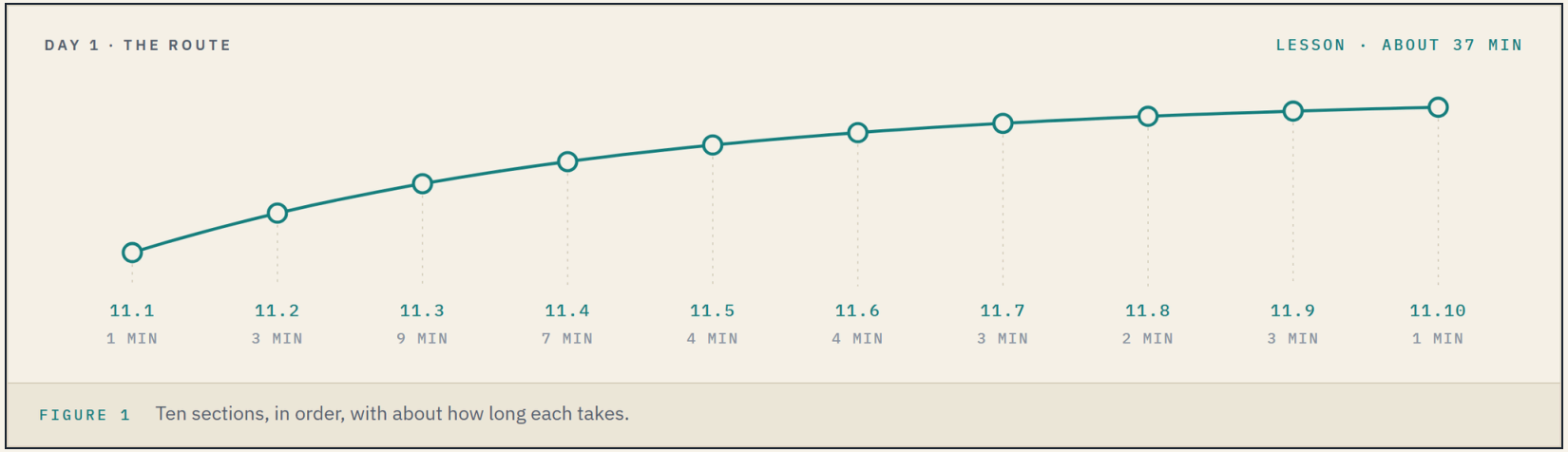


FIGURE 1 Ten sections, in order, with about how long each takes.

	SECTION	ABOUT
11.1	The Book and Two Libraries	1 min
11.2	The Anatomy of a Chart	3 min
11.3	Histogram	9 min
11.4	Box Plot	7 min
11.5	Bar Chart	4 min

	SECTION	ABOUT
11.6	Line Plot	4 min
11.7	Scatter Plot	3 min
11.8	Pair Plot	2 min
11.9	Heatmap	3 min
11.10	Choosing a Chart	1 min

Time. About 37 minutes for the lesson, then about 21 for the seven exercises.

11.1 The Book and Two Libraries

THE CODE

```
holdings = pd.read_csv(data_folder + 'holdings.csv')
ratings = pd.read_csv(data_folder + 'ratings.csv')
book = pd.merge(holdings, ratings, on='rating')
bucket_order = ratings.sort_values('score')['bucket'].unique().tolist()
print(book.shape)
print(bucket_order)
```

```
(200, 22)
['AAA', 'AA', 'A', 'BBB', 'BB', 'B', 'CCC']
```

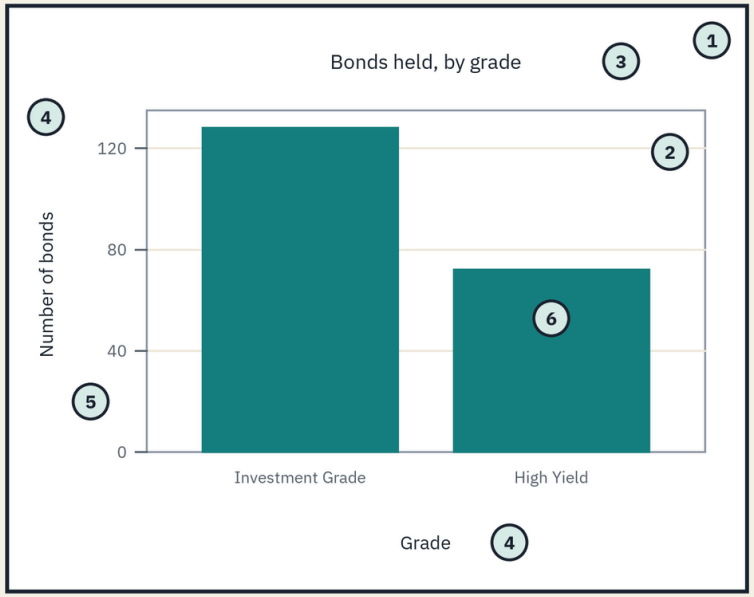
EXCEL SIDE BY SIDE

With the scale's score in column B and its bucket in column C, `=UNIQUE(SORTBY(C2:C19, B2:B19))` in Excel 2021 and Microsoft 365.

One table and one credit order, written once: every chart today draws from them.

11.2 The Anatomy of a Chart

The parts of a chart, and the line of code that makes each



1. `fig, ax = plt.subplots(figsize=(8, 4.5))` — the figure: Excel's Chart Area, sized by dragging a corner
2. `ax` — the axes, where the bars sit: Excel's Plot Area
3. `ax.set_title('Bonds held, by grade')` — the title: Excel's Chart Title
4. `ax.set_xlabel(), ax.set_ylabel()` — the axis labels, with the unit: Excel's Axis Titles
5. `ticks` — placed from the data: Excel's Format Axis
6. `ax.bar(grade_counts.index, grade_counts.values)` — High Yield is 72: =COUNTIF(V:V, "High Yield")

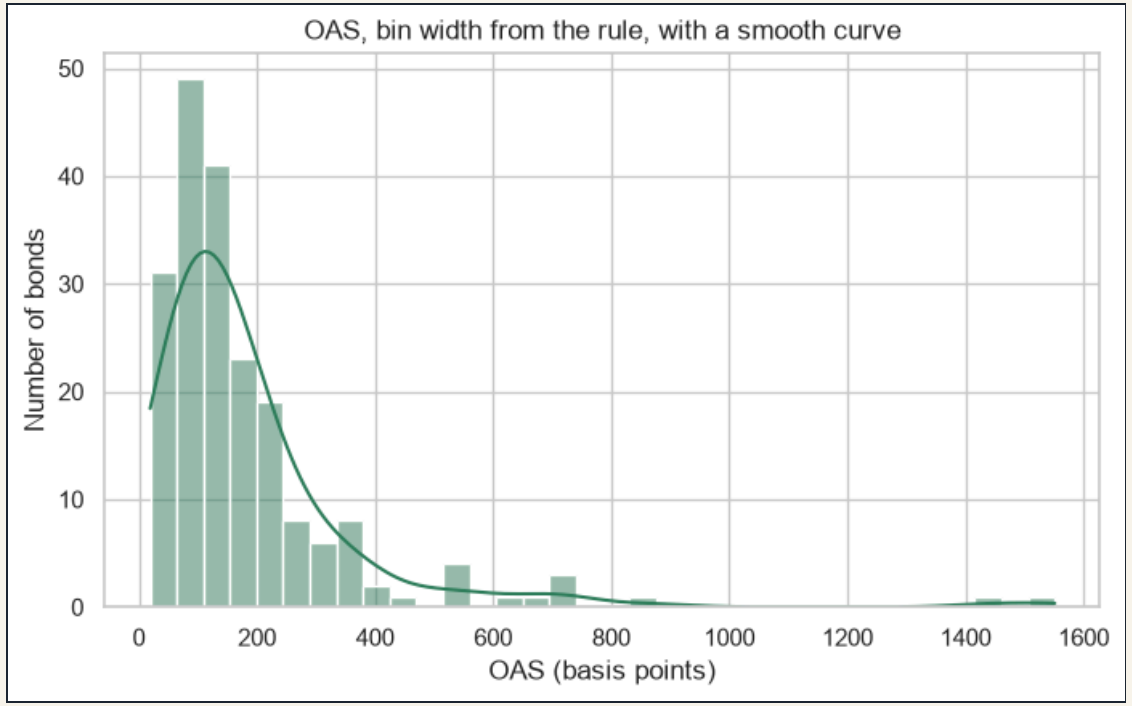
FIGURE 2 The grade bar chart with its six parts numbered, each with its matplotlib call and its Excel name.

MAKE, DRAW, LABEL

```
fig, ax = plt.subplots(figsize=(8, 4.5))
ax.bar(grade_counts.index, grade_counts.values)
ax.set_title('Bonds held, by grade')
ax.set_xlabel('Grade')
ax.set_ylabel('Number of bonds')
plt.show()
```

Make, draw, label, show: every chart today is built in that order.

11.3 Histogram



THE WIDTH FROM A RULE

```
iqr = q3 - q1
rule_width = 2 * iqr / len(book) ** (1 / 3)

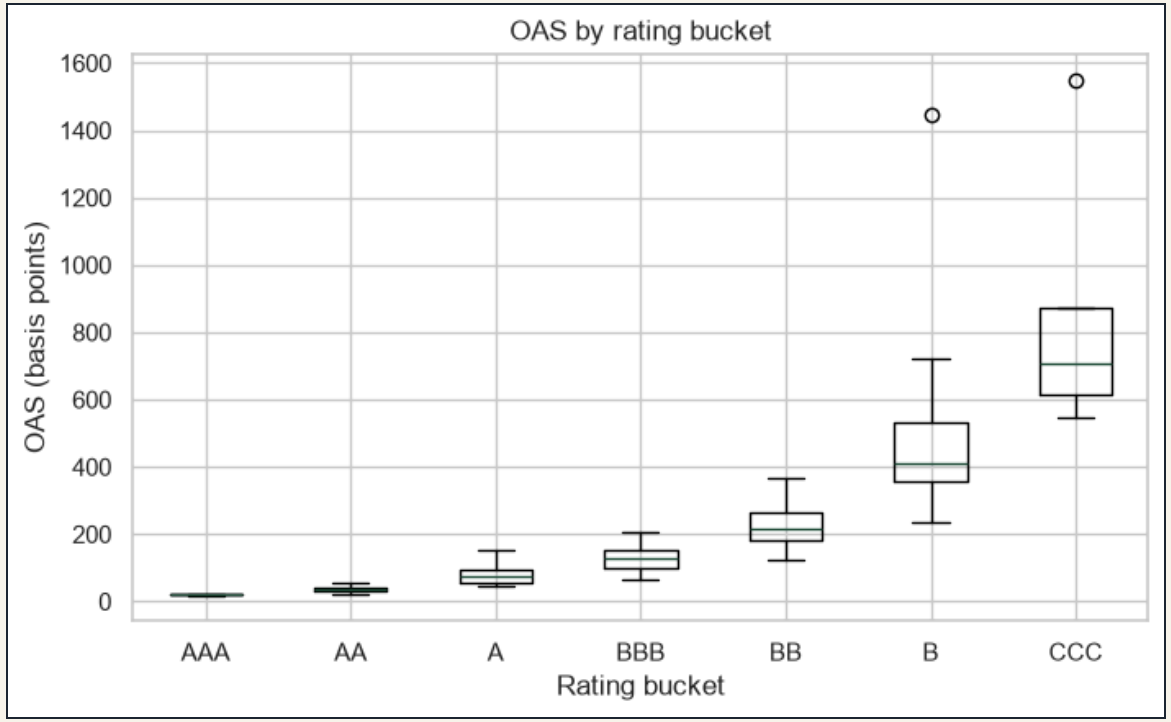
Width from the rule: 44.89 bp
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL
Histogram	`ax.hist(book['oas'], bins=20)`	Insert, Statistic Chart, Histogram
First of 5 bins	`counts[0]`, 173	`=COUNTIFS(M:M, ">=19", M:M, "<325.4")`
A note with an arrow	`ax.annotate(...)`	Insert, Shapes: an arrow and a Text Box

A histogram shows one column's shape; to compare groups, five numbers each are easier.

11.4 Box Plot



ONE BOX PER BUCKET

```
ax.boxplot(oas_by_bucket, tick_labels=bucket_order)
```

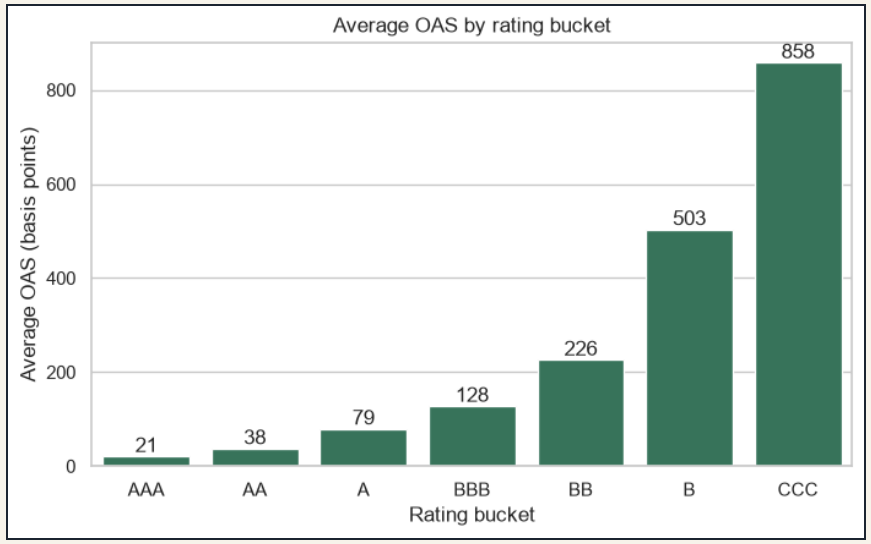
```
{'AAA': 2, 'AA': 8, 'A': 56, 'BBB': 62, 'BB': 49, 'B': 18, 'CCC': 5}
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL
Box plot	`ax.boxplot(book['oas'])`	Insert, Statistic Chart, Box and Whisker
Quartiles	`quantile(0.25)`, 81.75	`=QUARTILE.INC(M2:M201, 1)`, and 3 for 213
Log scale	`ax.set_yscale('log')`	Format Axis, Logarithmic scale

Count the bonds behind a box, and draw its histogram before trusting it. Now one number per group.

11.5 Bar Chart



CREDIT ORDER, VALUES ON THE BARS

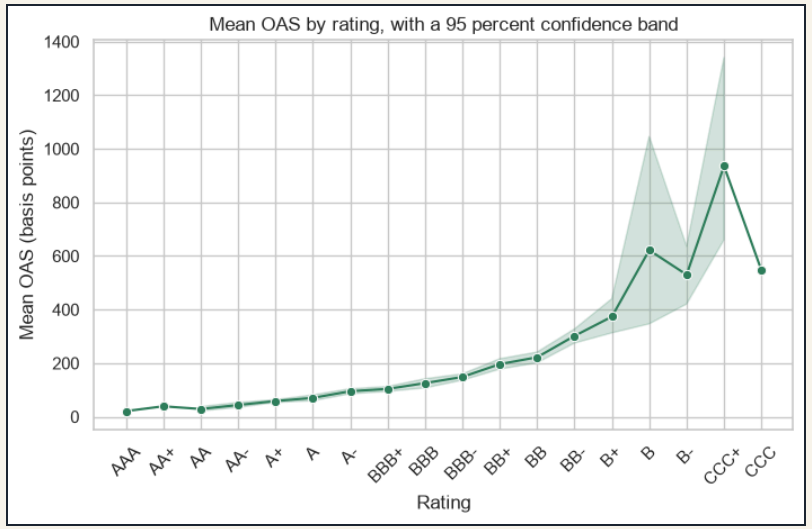
```
sns.barplot(data=book, x='bucket', y='oas', order=bucket_order, errorbar=None, ax=ax)
ax.bar_label(ax.containers[0], fmt='%.0f')
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL
One group's average	the BBB bar, 128	`=AVERAGEIFS(M:M, U:U, "BBB")`, 127.82
Upright or flat bars	`ax.bar()`, `ax.barh()`	Clustered Column, Clustered Bar
Tilted labels	`ax.tick_params(axis='x', rotation=45)`	Format Axis, Custom angle

No library knows credit order: pass it with `order=`.

11.6 Line Plot



A BAND FOR HOW SURE

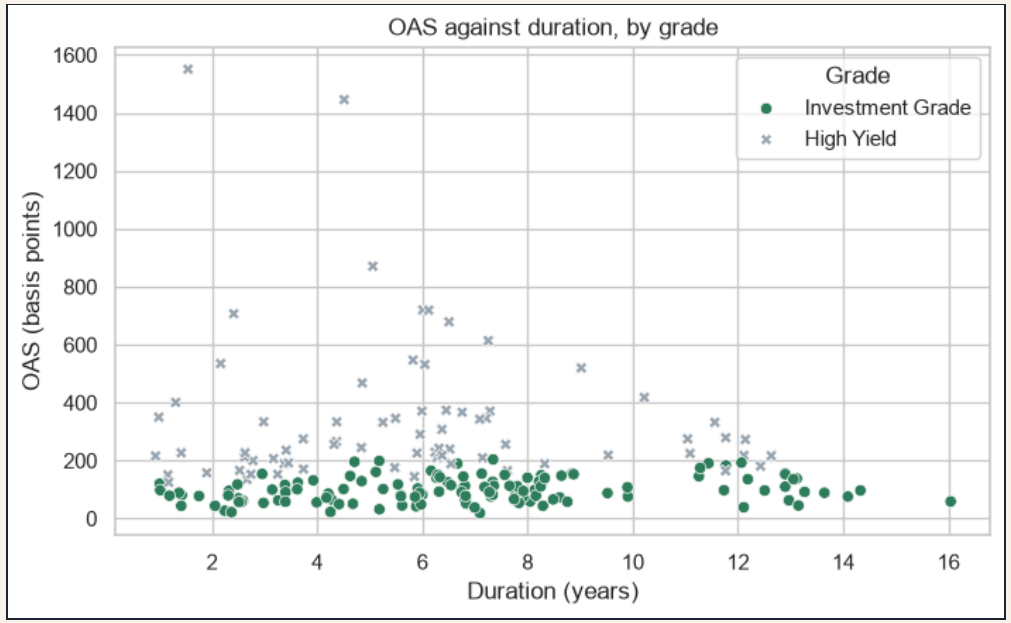
```
sns.lineplot(data=book, x='score', y='oas', errorbar=('ci', 95), seed=0, marker='o', ax=ax)
ax.set_xticks(ratings['score'], ratings['rating'], rotation=45)
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL
One group's median	`by_score.loc[10, 'median']`, 149	`=MEDIAN(IF(T2:T201=10, M2:M201))`, Ctrl+Shift+Enter before dynamic arrays
Line plot	`ax.plot(x, y, marker='o')`	Line with Markers

Dots for the data, a band for how sure the average is. Now two measures at once.

11.7 Scatter Plot



CORRELATION WITHIN EACH GRADE

```
print('Investment Grade:', round(ig['duration'].corr(ig['oas']), 2))
print('High Yield:', round(hy['duration'].corr(hy['oas']), 2))
```

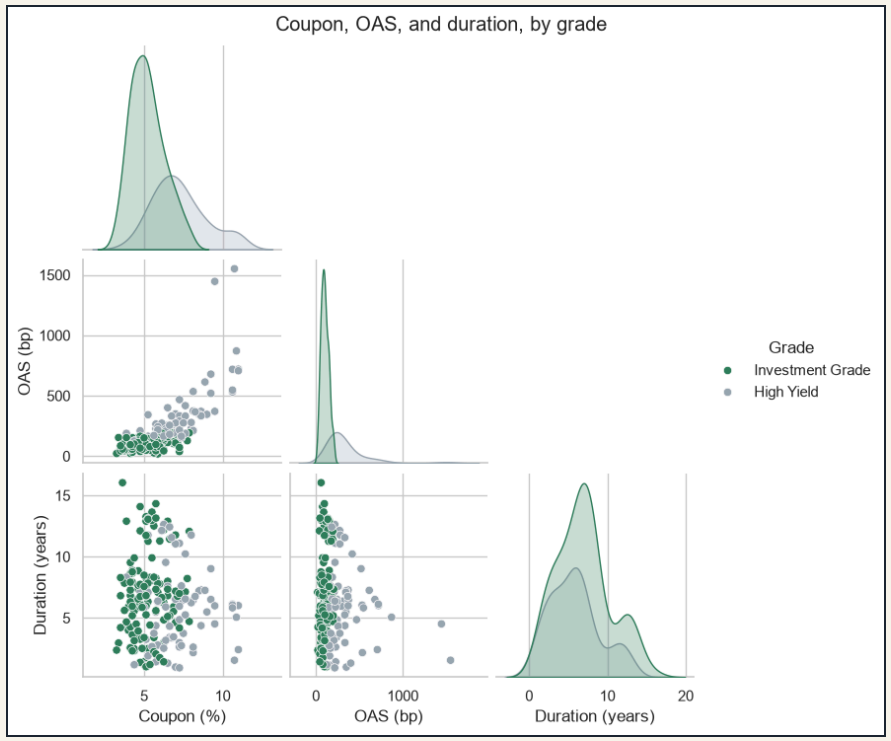
```
Investment Grade: 0.24
High Yield: -0.1
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL
Scatter	`ax.scatter(book['oas'], book['yield_pct'])`	Insert, Scatter; the first column goes on x
Correlation	`book['oas'].corr(book['yield_pct'])`	`=CORREL(M2:M201, L2:L201)`, 0.99

One number for the whole book can describe neither group. Now every pair at once.

11.8 Pair Plot



EACH PAIR ONCE, BY GRADE

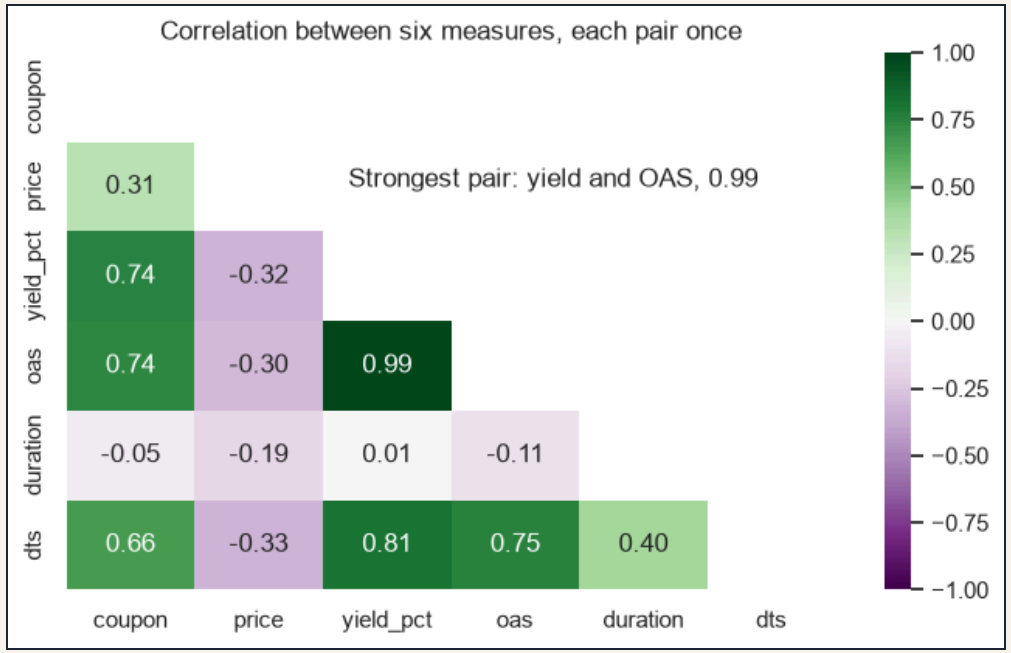
```
grid = sns.pairplot(pair_grade, vars=['Coupon (%)', 'OAS (bp)', 'Duration (years)'],
                    hue='Grade', corner=True, height=2.4)
```

EVERY PAIR OF FOUR COLUMNS

```
pair_data = book[['coupon', 'price', 'oas', 'duration']].rename(columns={
grid = sns.pairplot(pair_data, height=1.8)
```

A pair plot is a first look that picks the charts worth drawing. Now one number per pair.

11.9 Heatmap



MASK THE MIRROR HALF

```
upper_half = np.triu(np.ones_like(corr_table, dtype=bool))
sns.heatmap(corr_table, mask=upper_half, annot=True, fmt='.2f', cmap='PRGn',
            vmin=-1, vmax=1, ax=ax)
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL
Correlations	`corr()`	`=CORREL(M2:M201, N2:N201)`; for the table, the Analysis ToolPak's Correlation
Colour behind numbers	`sns.heatmap(table, annot=True)`	Conditional Formatting, Color Scales

The eye finds the dark cells before it reads a figure.

11.10 Choosing a Chart

The question picks the chart

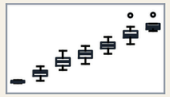
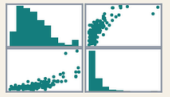
The question	Today's chart	Its name	matplotlib, seaborn	Excel
One column's spread		**Histogram**	`ax.hist` `sns.histplot`	Histogram
Groups compared		**Box plot**	`ax.boxplot`	Box and Whisker
One number per group		**Bar chart**	`ax.bar, ax.barh` `sns.barplot, sns.countplot`	Clustered Column or Bar
Change along an order		**Line plot**	`ax.plot` `sns.lineplot`	Line with Markers
Two columns together		**Scatter plot**	`ax.scatter` `sns.scatterplot`	Scatter
Every pair at a glance		**Pair plot**	`sns.pairplot`	a Scatter per pair
A table's large and small cells		**Heatmap**	`sns.heatmap`	Color Scales

FIGURE 4 Seven questions, each with today's chart, its matplotlib and seaborn calls, and its Excel chart.

THE HABITS

- Work out the numbers first, and check the chart against them.
- Title every chart, and label every axis with its unit.
- Put ratings in credit order; no library does.
- Know how many bonds stand behind each bar, box, or point.

You Can Now

- Make a figure and axes, draw on it, and label it with units.
- Read a histogram's and a box plot's shape, and set bins by count, width, or rule.
- Draw bars and lines in the order the categories need.
- Plot two measures by group, beside their correlation.
- Take a first look with a pair plot, and shade a table as a heatmap.

ASKED	CHART	ANSWER
The spread tail	histogram	164 below 250 bp, a tail to 1,551
Steps by rating	box plot	medians 21 to 707 bp
Where the money sits	bars	Materials 71.70 million
What moves together	heatmap	yield and OAS, 0.99

NEXT
WEEK 3 · DAY 2

More Visualization

A line through a scatter with its slope, the bonds behind every shape, charts that answer the mouse, and a chart finished for the meeting.